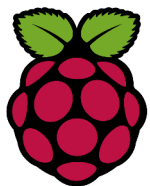


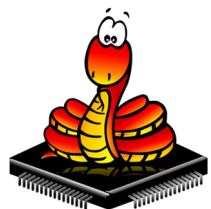


Le simulateur WOKWI

Atelier Raspi

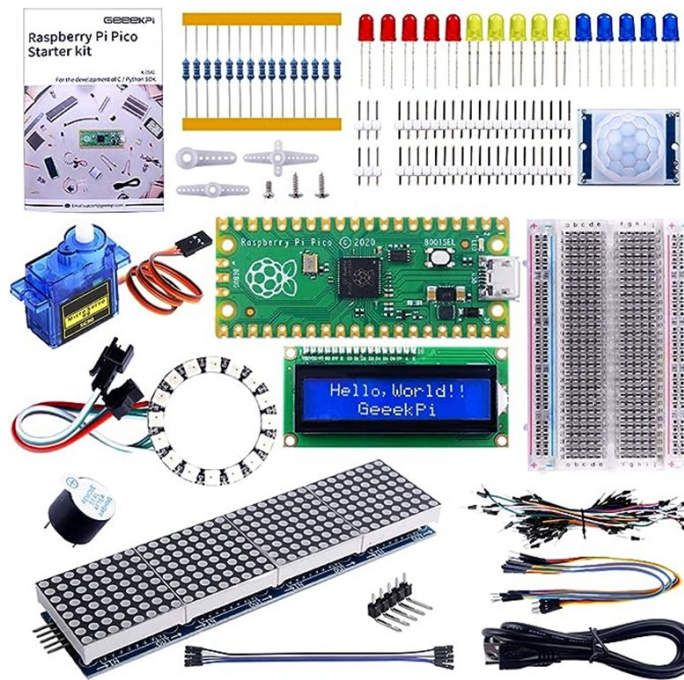


Logo du Raspberry Pico



Logo du MicroPython

L'atelier a pour valeurs, le partage, l'aide, la formation, le faire et construire ensemble à partir de l'expérience des participants



🚀 C'est quoi Wokwi ?

Wokwi, c'est un laboratoire d'électronique dans ton ordinateur.

Au lieu d'acheter des fils, des ampoules, des cartes électroniques et de risquer de tout casser...

👉 tu peux **tout tester directement** sur l'écran !

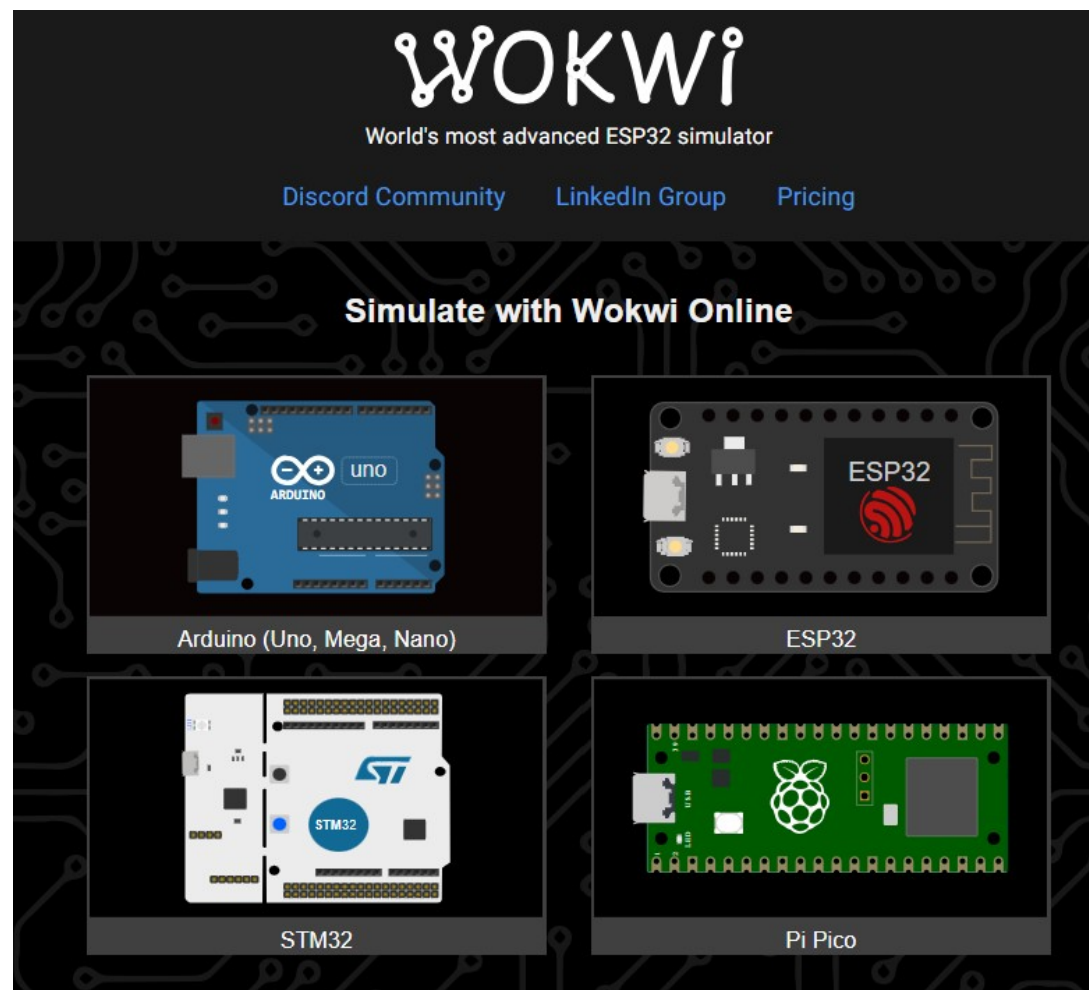
C'est un site internet qui permet de **programmer et construire des projets électroniques virtuellement**.

💡 Ça sert à quoi ?

Avec Wokwi, on peut :

- 💡 Allumer et éteindre des LED
- 🎵 Faire jouer des sons avec un buzzer
- 🌡️ Mesurer la température avec un capteur
- 🤖 Programmer des petits robots
- 💻 Apprendre à coder avec Arduino ou MicroPython

C'est comme un **jeu de construction + du code**.



WOKWi
World's most advanced ESP32 simulator

[Discord Community](#) [LinkedIn Group](#) [Pricing](#)

Simulate with Wokwi Online

Arduino (Uno, Mega, Nano)

ESP32

STM32

Pi Pico

https://wokwi.com

← ↻ 🔒 https://wokwi.com

🔍 A 🌟 ⚙️ 👤 Mettre à jour ... 🗨️ Conversation

WOKWI

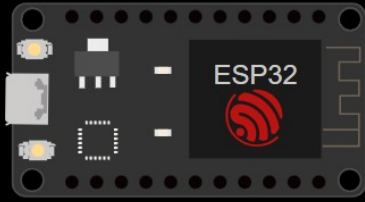
World's most advanced ESP32 simulator

[Discord Community](#) [LinkedIn Group](#) [Pricing](#)

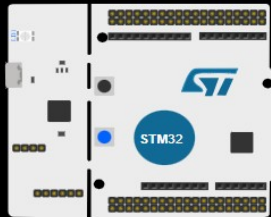
Simulate with Wokwi Online



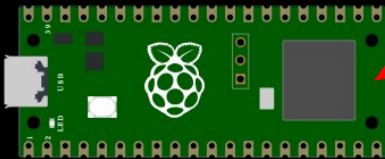
Arduino (Uno, Mega, Nano)



ESP32



STM32



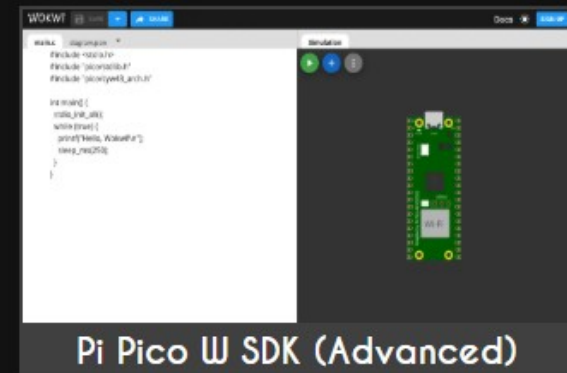
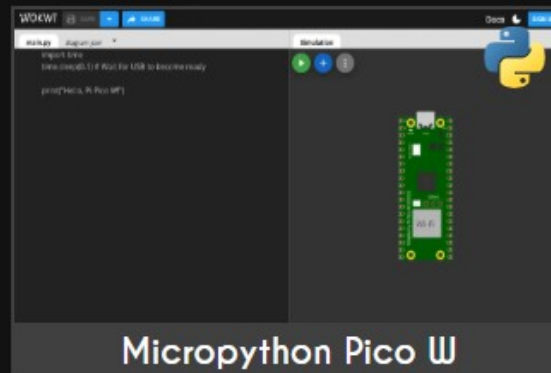
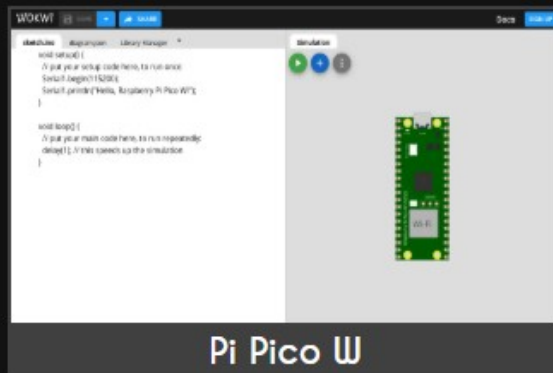
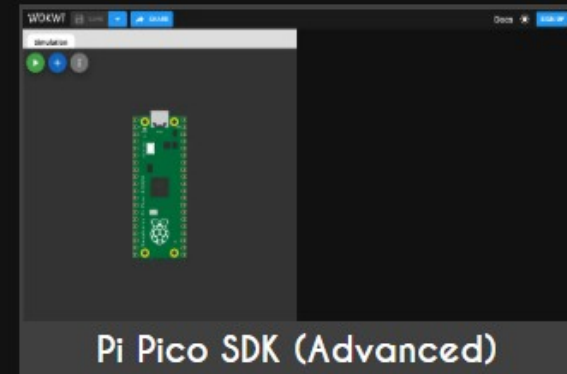
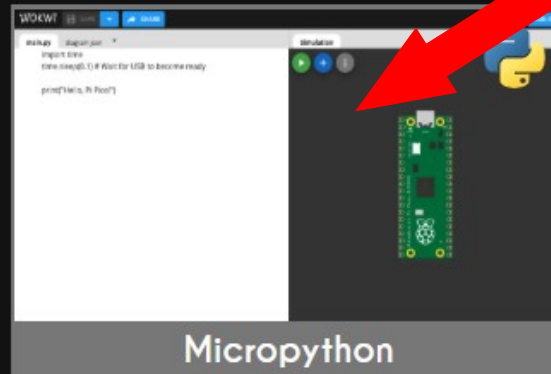
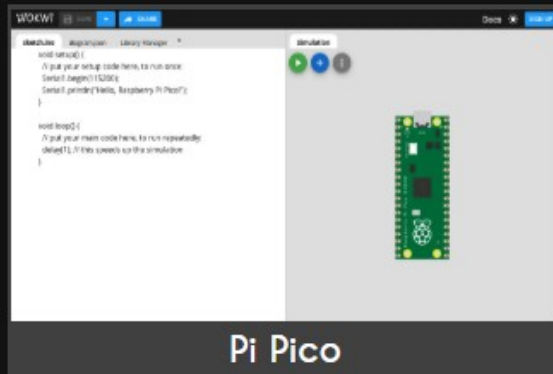
Pi Pico

Wokwi Pro

On sélectionne le microcontrôleur

On sélectionne un exemple simple
qui utilise le langage Micropython

Starter Templates



← ↻ <https://wokwi.com/projects/new/micropython-pi-pico> ㊚ A ☆ ☆ Mettre à jour ... Conversation

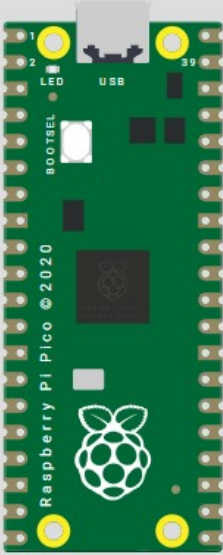
WOKWI SAUVEGARDER PARTAGER Docs

main.py diagram.json

```
1 import time
2 time.sleep(0.1) # Wait for USB to become ready
3
4 print("Hello, Pi Pico!")
5
```

Simulation

▶ + ⋮



Example simple de code

On lance la simulation

← ↻ <https://wokwi.com/projects/new/micropython-pi-pico> 日本語 A ☆ ☆ Mettre à jour ... Conversation

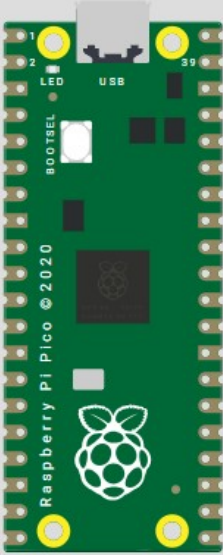
WOKWI SAUVEGARDER PARTAGER Docs

main.py diagram.json

```
1 import time
2 time.sleep(0.1) # Wait for USB to become ready
3
4 print("Hello, Pi Pico!")
5
```

Simulation

▶ + ⋮



Raspberry Pi Pico © 2020

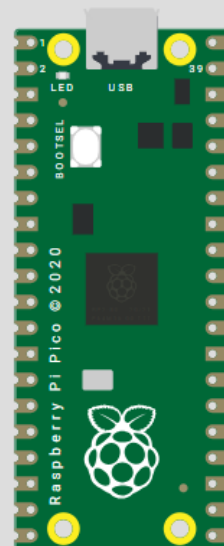
The image shows the Wokwi web interface for a Raspberry Pi Pico project. On the left, a code editor displays a Python script in 'main.py' that imports the 'time' module, sleeps for 0.1 seconds, and prints 'Hello, Pi Pico!'. On the right, a simulation panel shows a digital representation of the Raspberry Pi Pico board. Above the board, there are three circular buttons: a green play button, a blue plus button, and a grey three-dot menu button. A large red arrow points from the text 'On lance la simulation' to the green play button. The top of the interface includes a browser address bar, navigation icons, and a 'Docs' link.

main.py diagram.json

```
1 import time
2 time.sleep(0.1) # Wait for USB to become ready
3
4 print("Hello, Pi Pico!")
5
```

Simulation

🕒 00:19.012 🔁 100%



```
Hello, Pi Pico!
MicroPython v1.24.1 on 2024-11-29; Raspberry Pi Pico with RP2040
Type "help()" for more information.
>>>
```

WOKWI

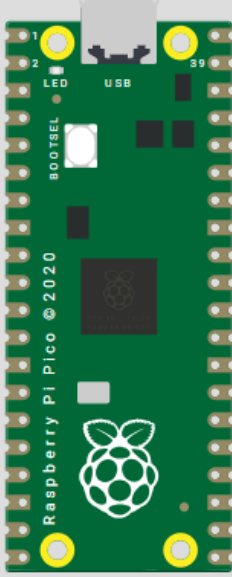
SAUVEGARDER PARTAGER

Docs

main.py **diagram.json**

```
1 {
2   "version": 1,
3   "author": "Anonymous maker",
4   "editor": "wokwi",
5   "parts": [
6     {
7       "type": "wokwi-pi-pico",
8       "id": "pico",
9       "top": -118.35,
10      "left": -54,
11      "attrs": { "env": "micropython-20241129-v1.24.1" }
12    }
13  ],
14  "connections": [],
15  "dependencies": {}
16 }
```

Simulation



The image shows a screenshot of the Wokwi web interface. The left pane displays a code editor with a file named 'diagram.json' selected, which contains a JSON configuration for a Raspberry Pi Pico simulation. The right pane shows a 3D simulation of the Raspberry Pi Pico board with control buttons (play, add, menu) above it. The browser address bar shows the URL 'https://wokwi.com/projects/new/micropython-pi-pico'.

Le fichier .json décrit les branchements de notre montage

On peut rajouter des composants

WOKWI

SAUVEGARDER PARTAGER

main.py diagram.json

```
1 {
2   "version": 1,
3   "author": "Anonymous maker",
4   "editor": "wokwi",
5   "parts": [
6     {
7       "type": "wokwi-pi-pico",
8       "id": "pico",
9       "top": -118.35,
10      "left": -54,
11      "attrs": { "env": "micropython-20241129-v1.24.1" },
12    }
13  ],
14  "connections": [],
15  "dependencies": {}
16 }
```

Basic

- LED
- Bouton poussoir
- Pushbutton 6mm
- Résistance

Display

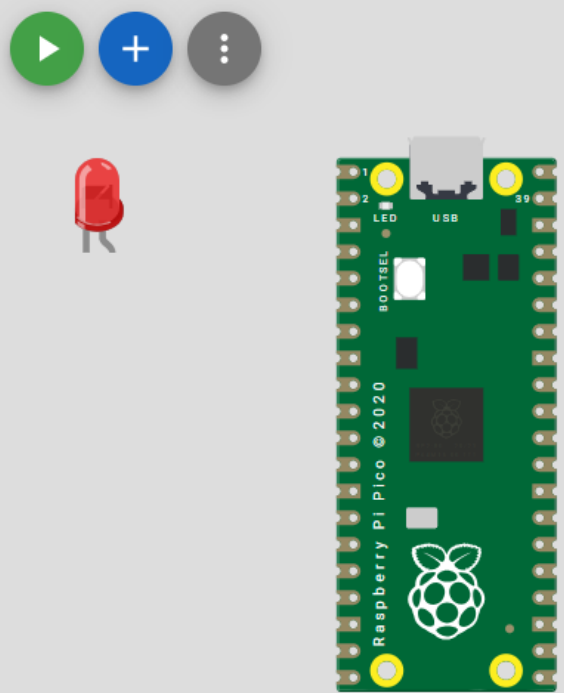
- LED RGB
- Ecran SSD1306 OLED
- Grove SH1107 - OLED Display
- LCD 16x2

WOKWI SAUVEGARDER PARTAGER Docs

main.py diagram.json

```
1 {
2   "version": 1,
3   "author": "Anonymous maker",
4   "editor": "wokwi",
5   "parts": [
6     {
7       "type": "wokwi-pi-pico",
8       "id": "pico",
9       "top": -118.35,
10      "left": -54,
11      "attrs": { "env": "micropython-20241129-v1.24.1" }
12    },
13    {
14      "type": "wokwi-led",
15      "id": "led1",
16      "top": -119.30867767333984,
17      "left": -160.3144989013672,
18      "attrs": { "color": "red" }
19    }
20  ],
21  "connections": [],
22  "dependencies": {}
23 }
```

Simulation

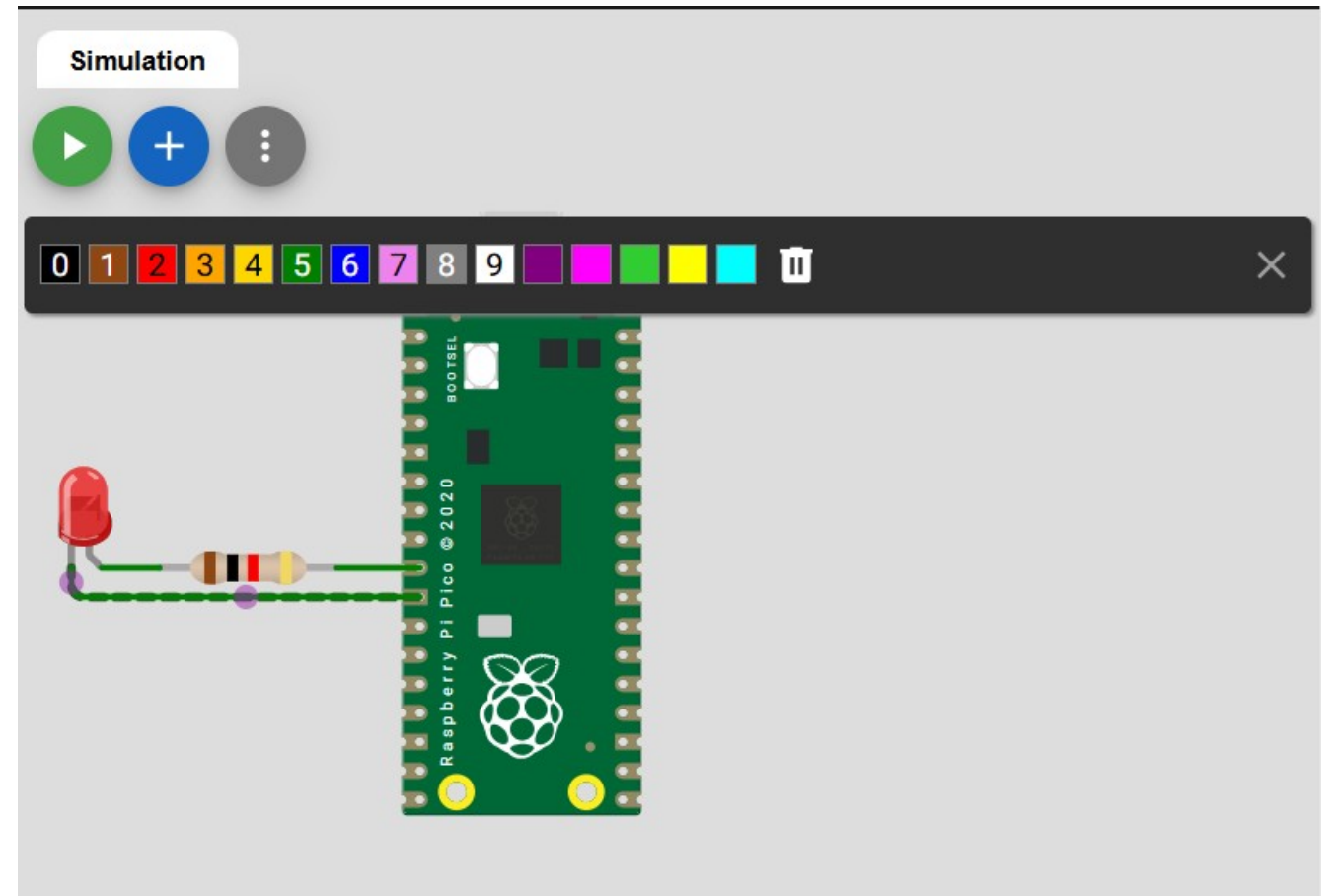
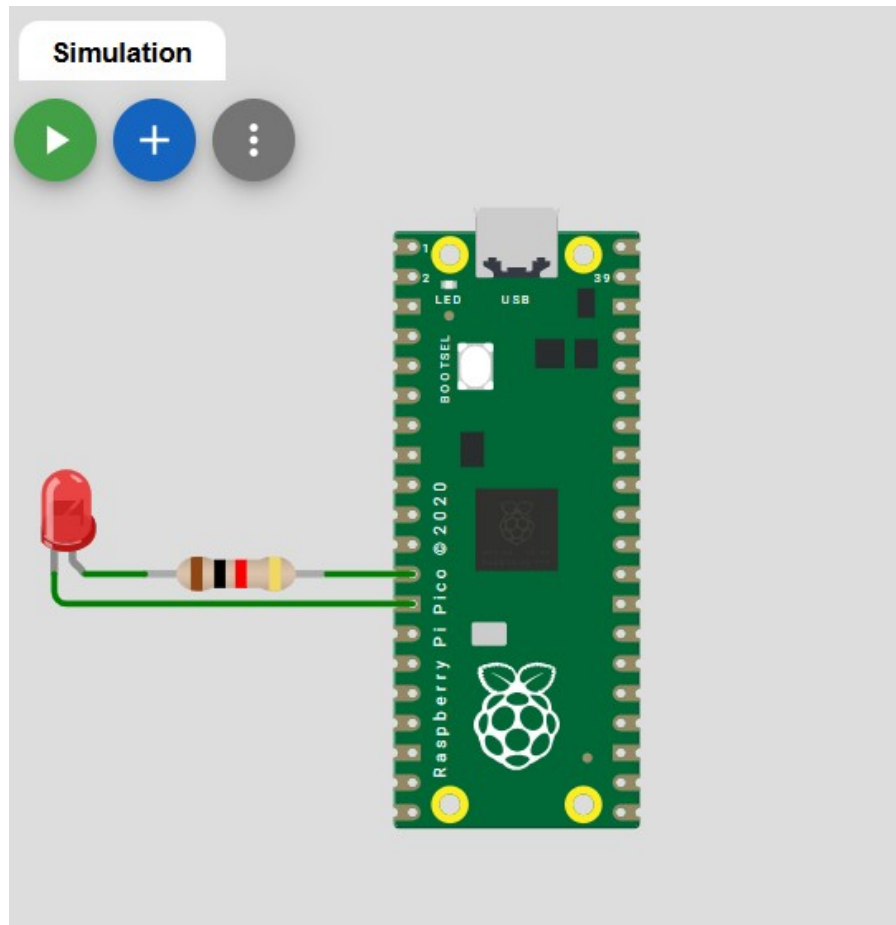


La led ajoutée apparaît dans le fichier json

On connecte la led au pico, on adapte la couleur des fils

Code couleur des câbles :

- Vcc : Rouge / Orange
- Gnd : Noir / Marron
- Données : Vert / Bleu / Jaune / Violet



WOKWI

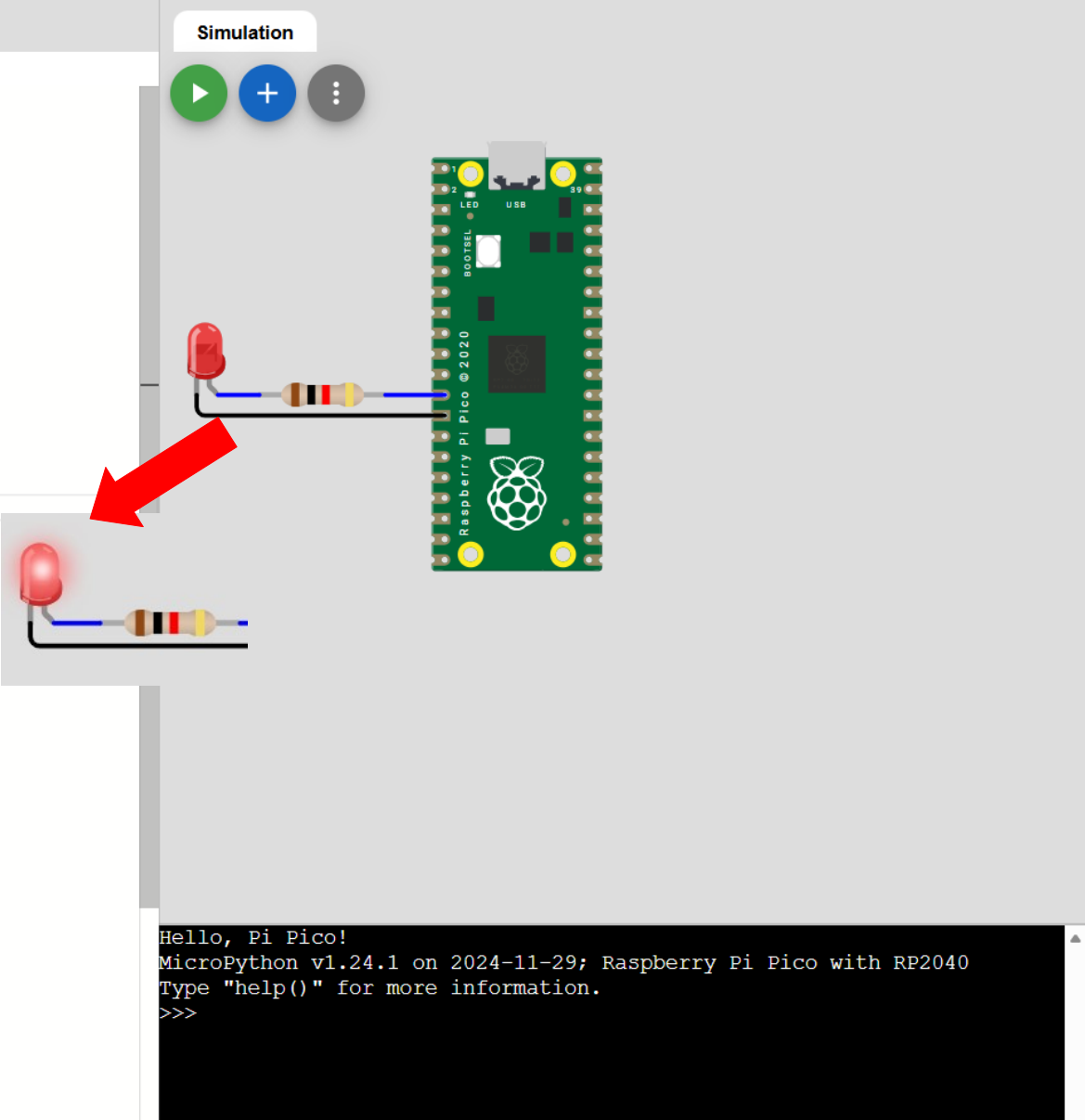
SAUVEGARDER PARTAGER

Docs

main.py • diagram.json •

```
1 from machine import Pin
2 import time
3 time.sleep(0.1) # Wait for USB to become ready
4
5 print("Hello, Pi Pico!")
6
7 LED = Pin(9, Pin.OUT)
8 #On donne la valeur 1 à la sortie pour allumer la LED
9 LED.value(1)
10 #On lui dit de rester allumée pendant 3 secondes
11 time.sleep(3)
12 #On lui dit de s'eteindre
13 LED.value(0)
14 #On lui dit de rester éteinte pendant 2 secondes
15 time.sleep(2)
16
```

Simulation

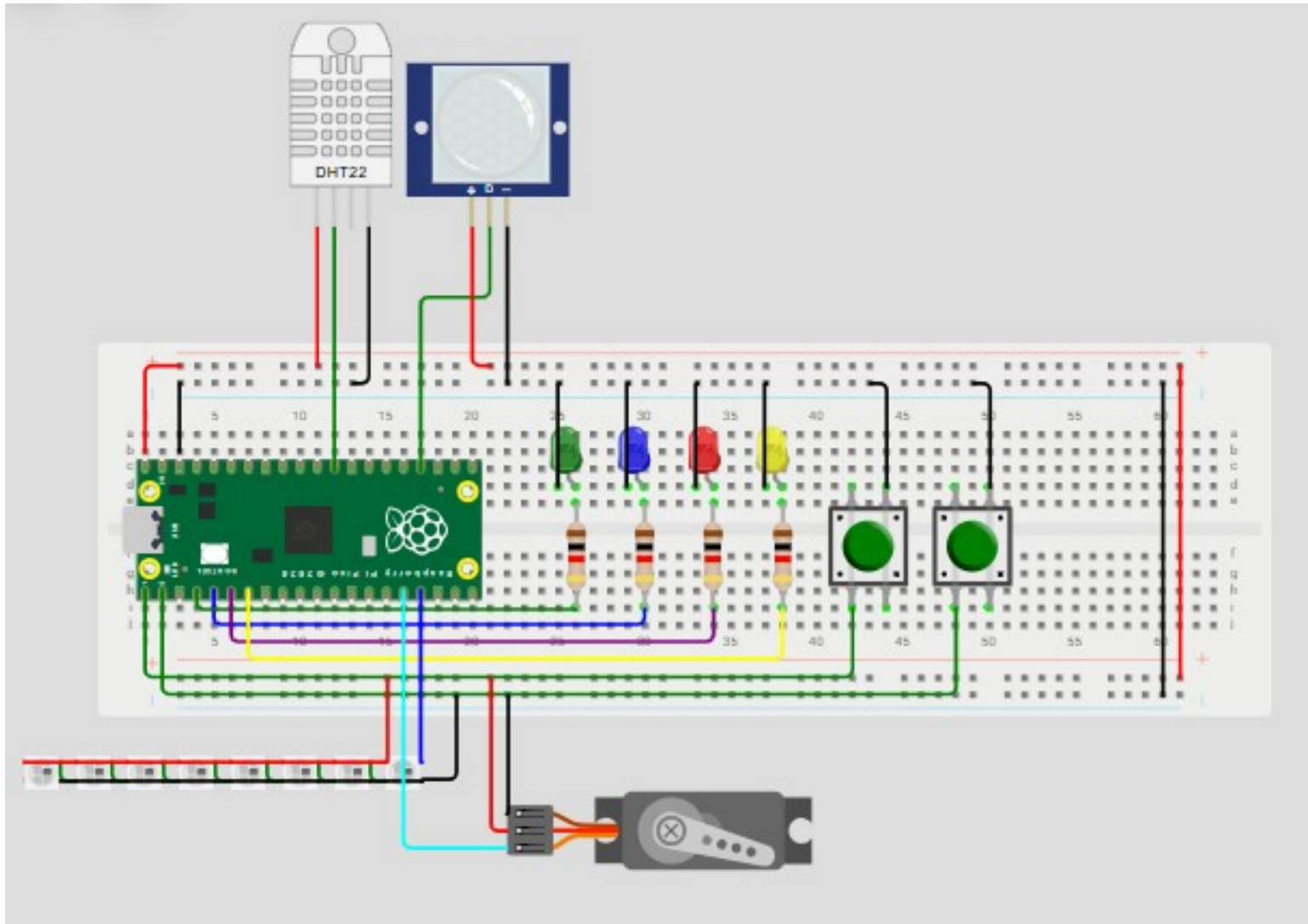


Simulation

Hello, Pi Pico!
MicroPython v1.24.1 on 2024-11-29; Raspberry Pi Pico with RP2040
Type "help()" for more information.
>>>

On peut maintenant écrire le code
et lancer la simulation.
On voit que la led s'allume.

Wokwi est utilisé pour la préparation des cours !



Composant	GPIO
Led Verte	2
Led Bleue	3
Led Rouge	4
Led Jaune	5
DHT11/ DHT22	22
PIR	18
Ruban Neopixel	13
BP1	0
BP2	1
Servo	12

Code couleur des câbles :

- Vcc : Rouge / Orange
- Gnd : Noir / Marron
- Données : Vert / Bleu / Jaune / Violet

WOKWI

SAUVEGARDER

PARTAGER

Mini Serre
by laurence

Docs

diagram.json

neopixel.py

main.py

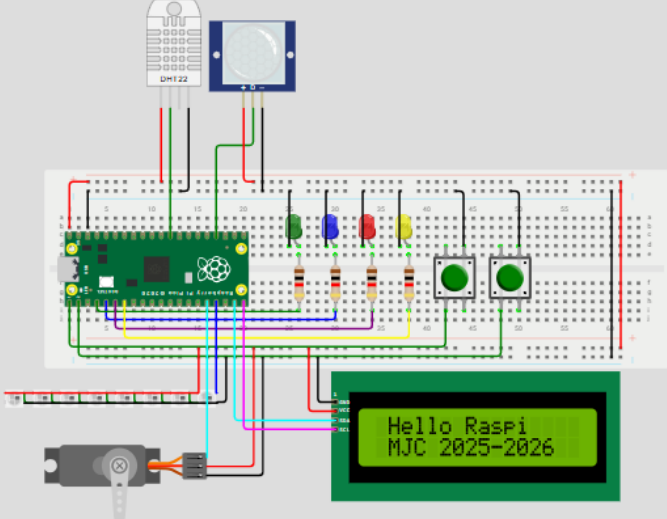
lcd_api.py

machine_i2c_lcd.py

```
1 from machine import Pin, I2C
2 from time import sleep
3 from machine_i2c_lcd import I2cLcd
4
5
6 i2c = I2C(1, scl=Pin(15), sda=Pin(14), freq=400000)
7
8 addr = i2c.scan()[0]
9 print(hex(addr))
10
11 lcd = I2cLcd(i2c, addr, 2, 16)
12
13 while True:
14     lcd.putstr(" Hello Raspi \n")
15     lcd.putstr(" MJC 2025-2026 \n")
16     sleep(1)
17     #lcd.clear()
18
19
```

Simulation

00:02.249 99%



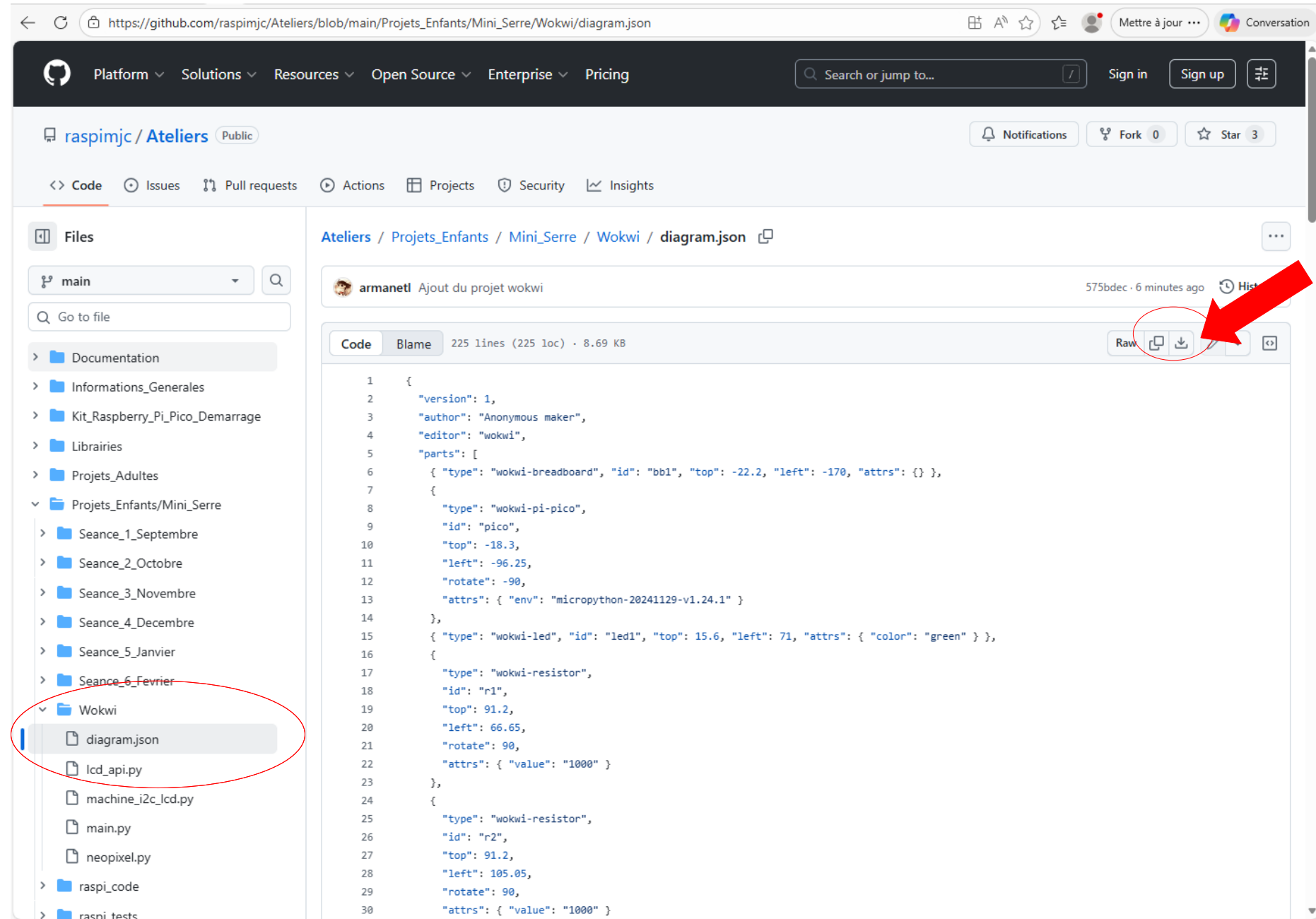
0x27

Exemple de la séance sur l'écran LCD

A noter : les librairies (type Neopixel) doivent être importées
En tant que fichier supplémentaire.
N'hésitez pas à nous demander de l'aide lors des séances à la MJC)

Astuce :
Le fichier json peut etre
copié depuis le github de
la MJC et collé dans
votre projet Wokwi !

Vous aurez directement
le montage du cours !



The screenshot shows the GitHub interface for the repository `raspimjc / Ateliers`. The file `diagram.json` is selected in the file explorer on the left. The main content area displays the JSON code for the file. A red circle highlights the file `diagram.json` in the file explorer, and another red circle highlights the download icon in the file header, with a red arrow pointing to it.

Files

- main
- Go to file
- Documentation
- Informations_Generales
- Kit_Raspberry_Pi_Pico_Demarrage
- Librairies
- Projets_Adultes
- Projets_Enfants/Mini_Serre
 - Seance_1_Septembre
 - Seance_2_Octobre
 - Seance_3_Novembre
 - Seance_4_Decembre
 - Seance_5_Janvier
 - Seance_6_Fevrier
 - Wokwi
 - diagram.json**
 - lcd_api.py
 - machine_i2c_lcd.py
 - main.py
 - neopixel.py
 - raspi_code
 - raspi_tests

Ateliers / Projets_Enfants / Mini_Serre / Wokwi / diagram.json

armanetl Ajout du projet wokwi 575bdec · 6 minutes ago

Code Blame 225 lines (225 loc) · 8.69 KB

```
1 {
2   "version": 1,
3   "author": "Anonymous maker",
4   "editor": "wokwi",
5   "parts": [
6     { "type": "wokwi-breadboard", "id": "bb1", "top": -22.2, "left": -170, "attrs": {} },
7     {
8       "type": "wokwi-pi-pico",
9       "id": "pico",
10      "top": -18.3,
11      "left": -96.25,
12      "rotate": -90,
13      "attrs": { "env": "micropython-20241129-v1.24.1" }
14    },
15     { "type": "wokwi-led", "id": "led1", "top": 15.6, "left": 71, "attrs": { "color": "green" } },
16     {
17       "type": "wokwi-resistor",
18       "id": "r1",
19       "top": 91.2,
20       "left": 66.65,
21       "rotate": 90,
22       "attrs": { "value": "1000" }
23     },
24     {
25       "type": "wokwi-resistor",
26       "id": "r2",
27       "top": 91.2,
28       "left": 105.05,
29       "rotate": 90,
30       "attrs": { "value": "1000" }
```